

MACHINE LEARNING FUNDAMENTALS SERIES

Neural Networks Pt. 4: Multiple Inputs & Outputs

Visualizing 3D Crinkled Surfaces for Multi-Class Iris Species Prediction

Source: StatQuest with Josh Starmer — Neural Networks Pt. 4: Multiple Inputs and Outputs

Crucial Pedagogical Declaration

Pre-Optimized Network Parameters: The neural network weights and biases shown in this document are **pre-optimized** via Backpropagation. This document focuses on the **geometric intuition** of how multi-input, multi-output networks construct 3D surfaces to classify data, skipping tedious step-by-step arithmetic to highlight the core concepts.

1. Introduction: Dimensionality of Inputs and Outputs

In single-input neural networks, predictions reside on a 2D plane where the x-axis represents the single input (x) and the y-axis represents the output ($\hat{y}$). Adding inputs and outputs expands the geometric space:

- **1 Input (x), 1 Output ($\hat{y}$):** Generates a **2D curve / squiggle** on a 2D Cartesian plane (x vs. $\hat{y}$).
- **2 Inputs (x_1, x_2), 1 Output ($\hat{y}$):** Generates a **3D surface / crinkled sheet** in a 3D space (x_1, x_2 vs. $\hat{y}$).
- **2 Inputs (x_1, x_2), 3 Outputs ($\hat{y}_1, \hat{y}_2, \hat{y}_3$):** Generates **three distinct 3D crinkled surfaces** simultaneously, one for each output class.

2. Minimalist Multi-Input, Multi-Output Network Topology

Consider the classic Iris classification problem: predicting Iris species (**Setosa**, **Versicolor**, **Virginica**) using two scaled flower measurements: **Petal Width** ($x_1 \in [0, 1]$) and **Sepal Width** ($x_2 \in [0, 1]$).

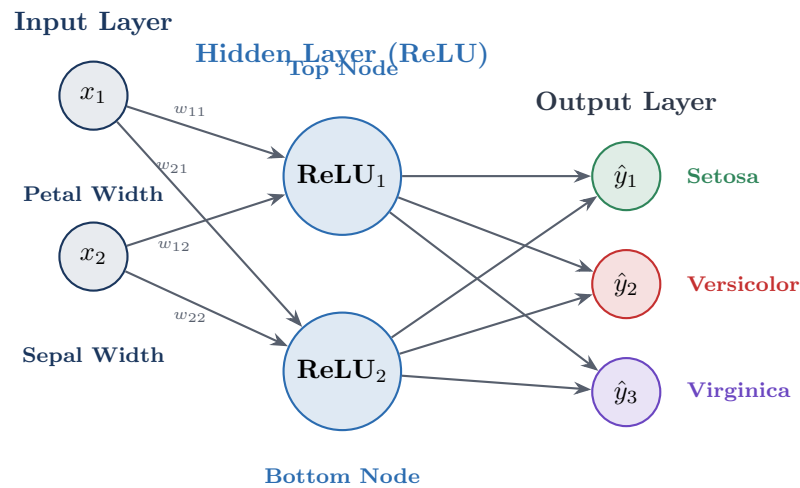


Figure 1: Minimalist Multi-Input Multi-Output Neural Network Topology (2 Inputs, 2 Hidden ReLU Nodes, 3 Output Species Classes). Weights and biases are pre-trained.

3. Geometric Mechanism: Constructing 3D Crinkled Surfaces

For two inputs (x_1, x_2), each hidden node evaluates a linear combination $z = w_1x_1 + w_2x_2 + b$ passed through the ReLU activation function $\text{ReLU}(z) = \max(0, z)$:

- Individual Bent Surfaces:** Each hidden node produces a 3D planar surface with a **straight fold/bend** along the line where $z = 0$.
- Output Scaling & Flipping:** Connections to an output node multiply the bent surfaces by output weights (stretching or lifting/flipping them upside down).
- Surface Summation:** The output node sums the scaled blue and orange bent surfaces and adds an output bias, fusing them into a **3D Crinkled Surface**.

4. 3D Visualizations of Output Crinkled Surfaces for 3 Species

Each output class receives a unique combination of output weights, carving out distinct 3D crinkled surfaces over the input domain $(x_1, x_2) \in [0, 1] \times [0, 1]$:

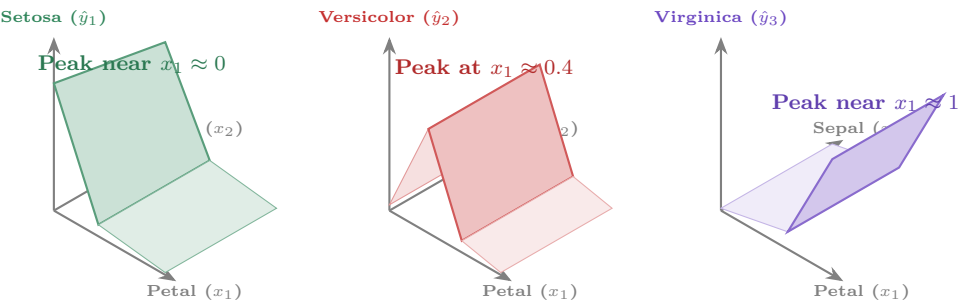


Figure 2: Isometric 3D Output Surface Representations for the Three Species. Each species achieves its maximum score in a distinct region of Petal Width (x_1).

5. Summary of 3D Output Characteristics

Table 1: Comparison of Output Surfaces Across Species Classes

Species Class	3D Surface Behavior	Peak Location	Final Decision Rule
Setosa ($\hat{y}_1$)	High near $x_1 = 0$, drops rapidly as $x_1 \rightarrow 1$.	Small Petal Width ($x_1 \approx 0$)	Maximum $\hat{y}$ score
Versicolor ($\hat{y}_2$)	Forms a ridge/mound in the middle region.	Medium Petal Width ($x_1 \approx 0.4$)	Maximum $\hat{y}$ score
Virginica ($\hat{y}_3$)	Low for $x_1 < 0.6$, rises sharply as $x_1 \rightarrow 1$.	Large Petal Width ($x_1 \approx 1$)	Maximum $\hat{y}$ score

Core Takeaway: Multi-Output Decision Making

By evaluating an input sample (x_1, x_2) across all three 3D crinkled surfaces simultaneously, the network produces a score vector $(\hat{y}_1, \hat{y}_2, \hat{y}_3)$. The predicted class is determined by selecting the highest output value via **ArgMax** (or after applying **Softmax** for probability normalization).